AI-Based Non-Contact Vital Sign Monitoring Using rPPG for Sustainable Healthcare

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Abstract— **The continuous and accurate measurement of physiological signs such as heart rate and respiratory rates plays a vital role in the early detection of possible health risks and in decision making within the healthcare setting. However, contact-based vital signs monitoring systems have been discovered to be intrusive and unpleasant, making them impractical for extensive use. An AI-based vital signs monitoring system using Remote Photoplethysmography (rPPG), which estimates physiological parameters from subtle variations in the facial skin color obtained using an ordinary RGB camera, is suggested in this article. It uses computer vision techniques for face detection and ROI selection together with a hybrid deep learning model, which incorporates both CNN and LSTM neural networks, in order to obtain spatial and temporal information from video streams. The technology has the ability to perform in real-time and is more robust against probable challenges such as motion and illumination changes. Furthermore, since the proposed system does not require any physical sensors, the technology proves to be hygienic and cost-effective.**

I. INTRODUCTION

Monitoring the parameters related to human physiology, such as heart rate, respiratory rate, and SpO2, is critical for early detection of diseases, clinical diagnosis, and managing patients' health status. Traditional monitoring equipment mainly uses contact sensing techniques like electrocardiogram, pulse oximeter, and wearables, which might be uncomfortable to use for long hours and do not lend themselves well to large-scale health care practices. However, due to growing requirements for hygiene and scalability in health care practices, researchers have focused on developing contactless monitoring methods.

Remote Photoplethysmography (rPPG) has been developed as an innovative technology for the measurement of vital signs without physical contact through the detection of skin color changes due to fluctuations in blood volume using ordinary RGB cameras. As opposed to other techniques that require sensors, rPPG is a reliable way to obtain physiological signals without touching the skin, which makes it extremely convenient for telemedicine, smart health care facilities, and constant surveillance of patients remotely. Remote detection of cardiovascular signals from video streams of the face has been proven feasible in numerous research projects through computer vision and signal processing methods. Initial research on remote pulse detection was carried out by Verkruysse *et al.* and Poh *et al.*

However, these improvements do not fully resolve problems associated with motion artifacts, variable lighting conditions, signal interference, and computational requirements, among others. This research proposes a novel method to overcome these drawbacks by developing a non-contact vital sign monitoring system based on computer vision and deep learning techniques for estimating vital signs. This system employs facial videos acquired by means of RGB cameras, where face detection and region of interest extraction are accomplished by applying the OpenCV library, and a hybrid CNN-LSTM network architecture that extracts spatial-temporal information from video sequences.

The model is trained using a massive dataset containing 293 files, out of which 193 files are videos and the rest are annotated in the form of JSON. The model is then coupled with a live streaming camera interface for the real-time prediction of heart rate, respiratory rate, and SpO_2 levels. The proposed technique is designed using Python programming along with TensorFlow, Keras, NumPy, Pandas, and Matplotlib libraries. This technique provides an economical and hygiene-friendly alternative to conventional health monitoring systems. Since there are no wearable devices involved, this study supports sustainability in healthcare. Remote health monitoring can be performed effectively without any physical contact with patients.

II. LITERATURE REVIEW

The growing attention towards the non-invasive healthcare monitoring system has led to innovations in the area of physiological signal acquisitions through cameras. One such technology is Remote Photoplethysmography (rPPG), which is effective in extracting heart rate, respiration rate, and oxygen saturation levels of individuals from video recordings of their face without needing any physical contact.

In an experiment conducted by Daniel McDuff, it was seen that small variations in the color and motion of faces through

*This work was not supported by any organization

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video camera could lead to determination of the individual's heart rate, respiratory rate, and oxygen saturation level. [1].

The first basis for developing remote photoplethysmography was laid down by Verkruysse, Svaasand, and Nelson, who could record the blood volume pulse remotely with ordinary digital cameras in ordinary lighting conditions. This study showed that pulse rate can be measured non-invasively even in natural lighting conditions, although motion artifacts were a major issue at this stage [3].

The concept was investigated even further by Poh, McDuff, and Picard through creating a machine-based non-contact method for measuring the pulse signal by utilizing video images of the facial region along with ICA. Although the method proved successful in capturing the heartbeat using the RGB facial image signal, it required very slight head movement. [4].

The problem of robustness against motion artifacts was tackled by Gerard de Haan and Vincent Jeanne using the CHROM approach, where chrominance based signal processing is used to discriminate between the pulse signal and motion artifacts. This method made considerable contributions in terms of increasing the robustness of pulse signal detection systems. [11].

A new technique by De Haan and van Leest that was resistant to movement and involved blood volume pulse signatures was developed, which showed significant improvement in signal detection over existing blind source separation techniques. [9].

Bin Lin *et al.* presented a deep learning model for combining video magnification and neural networks to amplify the fine skin motion corresponding to pulse variations prior to feature extraction. Although their technique increased heart rate measurement precision, it was more computationally demanding. [2].

In their study of motion-based detection of pulses, Balakrishnan, Durand, and Guttag considered the minute movement of the head resulting from the movement of blood during each heart beat. Their method was successful even under low-color conditions but was hindered by excess motion. [7].

Lu Niu *et al.* extended the application of rPPG from face-only detection to show that cardiovascular information could also be obtained from other parts of the body like the arm, hand, and leg. It demonstrated that several skin locations are suitable for remote physiological detection. [5].

In the study AMPLIFY, Abnousi *et al.* created a novel, non-invasive heart failure remote monitoring system based on Eulerian video magnification technology that enhances jugular venous pulses detected from the neck region through video footage. The study produced promising cardiology applications[6].

Moreover, the scientific rationale for photoplethysmography has been described in detail by John Allen, elucidating the principles of measuring the volume of blood pulses and their uses in cardiology [8]. Contributions to our understanding of the principles of light absorption and reflection by biological tissue have also been made by Ping Shi and Hongliu Yu [10].

Though considerable advancements have been made in non-contact vital signs monitoring, issues such as sensitivity to light variations, motion artifacts, instability of signals, and real-time processing remain unresolved. In order to overcome these issues, this paper proposes an AI-based system for rPPG using a combination of computer vision and a convolutional neural network (CNN)-long short-term memory (LSTM) deep learning approach.

III. Methodology

The system architecture adopts a contactless strategy to monitor the vital signs via Remote Photoplethysmography (rPPG) combined with deep learning algorithms. The overall framework includes the steps of video recording, face detection, extraction of the region of interest, signal processing, and estimating the physiological parameters via CNN-LSTM.

A. System Overview

The system workflow is as follows:

Video Input → Face Detection → ROI Extraction → rPPG Signal Extraction → CNN-LSTM Model → Heart Rate & Respiratory Rate Output

B. Dataset Description

In the experiment, a dataset is used that includes video recordings of faces together with their physiological signal labels, which serve as input data for training and validating the proposed rPPG-based system. There are 293 files included in the dataset, 193 files in `.mp4` format as videos and 99 files in `.json` format as labels. Video files represent recordings of faces under different external and internal conditions, whereas JSON files include physiological parameter labels like heart rate and respiratory rate.

Each video is a collection of frames that display minor variations in skin color due to changes in blood flow through the face region. Such variations serve as the basis of rPPG signal extraction. In addition, the dataset is constructed in a way that ensures the correspondence between videos and label files. This feature makes it possible to learn the relationship between inputs and outputs using supervised learning methods and predict vital parameters of subjects.

Moreover, the dataset contains videos taken under different lighting conditions, movements, and face positions, making the model more robust to external disturbances. Therefore, the proposed approach can provide good results when working in diverse unconstrained environments.

1) Dataset Summary:

- Total Files: 293
- Total Video Files: 193
- Total Label Files: 99
- Data Format: Video (`.mp4`) and Labels (`.json`)

TABLE I

SUMMARY OF LITERATURE SURVEY ON NON-CONTACT VITAL SIGN MONITORING USING rPPG

Author(s)	**Methodology Used**	**Findings**	**Drawbacks**
Gerard de Haan and Vincent Jeanne [11]	CHROM-based chrominance signal processing for facial pulse extraction	Improved pulse estimation accuracy by separating pulse signals from motion noise	Affected by severe illumination changes and excessive motion
Bin Lin *et al.* [2]	Deep learning with video magnification and neural networks	Enhanced subtle pulse variations and improved heart rate accuracy	High computational complexity limits real-time use
Ming-Zher Poh, Daniel McDuff, Rosalind Picard [4]	ICA-based RGB facial video signal decomposition	Accurate non-contact heart rate extraction using webcams	Requires stable face position; sensitive to motion
Daniel McDuff *et al.* [1]	Facial color and motion analysis through camera systems	Enabled estimation of heart rate, respiration, and SpO_2	Sensitive to lighting changes and subject movement
Gerard de Haan and van Leest [9]	Motion-resistant blood volume pulse signature analysis	Improved pulse reliability during movement	Limited under extreme motion conditions
Wim Verkruysse *et al.* [3]	Ambient-light remote photoplethysmographic imaging	Established foundation for remote pulse detection	Highly sensitive to motion and illumination fluctuations
Guha Balakrishnan *et al.* [7]	Motion-based pulse detection using head tracking and PCA	Effective when color signals are weak	Accuracy decreases with excessive head movement
Lu Niu *et al.* [5]	Full-body cardiovascular sensing from multiple skin regions	Showed arms, hands, and legs also provide useful signals	Requires complex region tracking
Freddy Abnousi *et al.* [6]	Eulerian video magnification for neck pulse monitoring	Strong potential in telemedicine cardiovascular monitoring	Needs further clinical validation
John Allen [8]	Review of traditional photoplethysmography principles	Provided theoretical foundation for blood pulse measurement	Focused on contact-based systems only
Ping Shi and Hongliu Yu [10]	Optical signal acquisition principles in biological tissues	Supported scientific basis of rPPG sensing	Mainly theoretical, lacks implementation model

C. Face Detection and ROI Extraction

The face detection process involves the use of ROI (Region of Interest) This classifier will detect the regions where faces exist in each video frame. It involves extraction of regions of interest (ROIs), including the forehead and cheeks, because these are areas where there exists good blood signal flow.

D. rPPG Signal Extraction

Remote Photoplethysmography (rPPG) captures subtle changes in skin color caused by blood circulation. These variations are extracted from RGB channels of the ROI.

Let the RGB signals be represented as:

$$R(t), \quad G(t), \quad B(t)$$

The combined signal is computed using a weighted combination of these channels:

$$S(t) = w_1 R(t) + w_2 G(t) + w_3 B(t)$$

where w_1, w_2, w_3 are weighting coefficients.

To improve robustness, the chrominance-based method (CHROM) is used:

$$X = 3R - 2G, \quad Y = 1.5R + G - 1.5B$$

$$S(t) = X - Y$$

The signal is then filtered using a bandpass filter to remove noise and retain frequencies corresponding to physiological signals.

E. Implementation Approach Based on Proposed Pipeline

The proposed system follows a structured pipeline as illustrated in the methodology diagram, consisting of sequential stages from video acquisition to real-time output display.

1) Video Acquisition: The system starts by recording videos of the face continuously through an RGB camera. These video recordings are broken down into frames at a constant frame rate (about 30 frames per second), thus providing enough temporal resolution for the capture of minute physiological changes due to blood flow.

2) Face Detection and ROI Tracking: This step in the detection process does not have any implementation using conventional classification algorithms. Instead, there is direct extraction of the Region of Interest (ROI) using predefined facial regions such as the forehead and cheeks from the video stream.

These areas exhibit high blood perfusion and hence are capable of producing rPPG signals. The ROI begins from the start of the video frames and continues tracking in future frames to maintain temporally consistent output signals.

ROI tracking involves just fixing the position of the area or applying the concept of consistency of frame to frame to minimize the computational overhead that arises from detecting the face.

Such a technique guarantees consistent capturing of the same facial area regardless of any motion made by the subject.

3) ROI (Region of Interest) Extraction: The face detected is used to select certain facial regions, such as the forehead and cheeks, which become the Regions of Interest

(ROI). This is because these facial regions have good blood perfusion as well as minimal obstruction and are therefore perfect for the extraction of the rPPG signal.

4) Preprocessing: Preprocessing is conducted on the ROI extracted frames through processes like normalization and noise removal. Any variance that comes about as a result of changes in lighting conditions or even small movements is eliminated through filtering techniques.

5) Feature Extraction (RGB-based Physiological Signal Analysis):

The proposed system analyzes subtle color variations in the facial skin using RGB channels extracted from the Region of Interest (ROI). Each frame is decomposed into its Red (R), Green (G), and Blue (B) components:

$$R(t), \quad G(t), \quad B(t)$$

These color channels capture physiological changes caused by blood circulation and oxygenation.

- **Green Channel (G):** The green channel is the most significant for heart rate estimation, as hemoglobin absorbs green light strongly. Variations in the green channel intensity correspond to blood volume changes during cardiac cycles. Therefore, heart rate (HR) is primarily derived from periodic fluctuations in the green channel.
- **Red Channel (R):** The red channel penetrates deeper into the skin and is sensitive to oxygenated blood levels. It is commonly used in combination with other channels for estimating oxygen saturation (SpO_2).
- **Blue Channel (B):** Blue is more responsive to changes at the surface level due to noise and light intensity fluctuations. Although it provides less information when used alone for physiological measurements, it can be combined effectively with R and G channels.

Heart Rate (HR): Heart rate is associated with high-frequency periodic variations in the green channel due to pulsatile blood flow.

Respiratory Rate (RR): Respiratory rate is based on slower fluctuations in the RGB signals, affected by the movement of the chest and small intensity variations in the face area.

Oxygen Saturation (SpO_2): The SpO_2 calculation is usually dependent upon the ratio between red and green channel signals because of their specific absorption properties for oxygenated and deoxygenated hemoglobin. While there are some theoretical formulas available, this paper proposes using the network to learn those properties.

6) CNN-LSTM Based Estimation: The feature vectors thus derived are fed into the proposed CNN-LSTM model. While the CNN part learns the spatial representation for each frame, the LSTM learns the temporal dependency among the sequences of frames. Unlike the conventional approach where the frequency components are calculated directly, the network will learn these components on its own through training.

7) Output Display: Prediction values for heart rate and breathing rate are obtained from the trained model in real-time. They are updated on an ongoing basis and presented to the user for real-time, contactless health tracking.

Alert and Notification System:

In order to make the system more applicable to practical use, an alerting module is included to alert users whenever the values of the vital signs recorded show any anomaly. In the implementation, the system keeps on monitoring the predicted rates of heartbeats and breathing breaths and compares them with normal ranges.

When the system detects that the estimated figures go out of range, it triggers an alert. An alert can be triggered when, for example, the heart rate is less than 60 beats per minute or more than 100. The alert message is then sent to the user through email.

Emailing in this case is done through a light-weight email server incorporated into the software application.

Relation to Theoretical Formulations: Despite the conventional methods involving calculations of physiological parameters using explicit algorithms like frequency estimation (FFT) and color signal analysis, the proposed algorithm does not explicitly use them to achieve the goal. Rather, the CNN-LSTM framework is trained to learn implicit representations based on input. The mathematical formulae described before constitute the theory behind the working of the model, which is replaced by the implemented system.

F. Deep Learning Model (CNN-LSTM)

A hybrid CNN-LSTM architecture is employed in the proposed system to model both spatial and temporal characteristics of physiological signals extracted from facial video data.

CNN Component: The Convolutional Neural Network (CNN) is used to extract spatial features from the input frames or ROI sequences. These features include subtle variations in pixel intensities and color distributions (particularly in RGB channels), which are associated with blood flow and physiological changes. The CNN automatically learns relevant feature maps such as edges, textures, and color patterns that contribute to rPPG signal representation.

LSTM Component: The Long Short-Term Memory (LSTM) network is used to capture temporal dependencies across consecutive frames. Since physiological signals such as heart rate and respiratory rate exhibit periodic behavior over time, the LSTM learns sequential patterns and temporal variations in the extracted features.

The internal functioning of an LSTM unit is governed by the following equations:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$$

$$h_t = o_t \cdot \tanh(C_t)$$

where f_t, i_t, and o_t represent the forget, input, and output gates respectively, and C_t denotes the cell state.

Implementation in the Proposed System: In the actual implementation, the CNN-LSTM model is constructed using deep learning frameworks such as TensorFlow/Keras. The model takes sequences of ROI-based frames or extracted feature vectors as input.

Unlike traditional signal processing approaches, the model does not explicitly compute physiological parameters using mathematical formulas. Instead, it learns the mapping between input video sequences and corresponding vital sign labels (heart rate and respiratory rate) through supervised learning.

The CNN layers automatically extract spatial features, which are then passed to LSTM layers that model temporal dynamics. Finally, fully connected layers produce the predicted values of heart rate and respiratory rate.

Note on Theoretical Formulation: Although the LSTM equations describe the internal operations of memory cells, these computations are not manually implemented in the project. Instead, they are handled internally by the deep learning framework. The inclusion of these equations provides a theoretical understanding of how temporal dependencies are learned within the model.

This data-driven approach enables the system to capture complex physiological patterns directly from video data, improving robustness against noise, motion artifacts, and illumination variations.

G. Heart Rate Estimation

The heart rate is calculated from the frequency of the rPPG signal.

$$HR = \frac{60 \times f_{peak}}{1}$$

where f_{peak} is the dominant frequency obtained using Fast Fourier Transform (FFT).

H. Respiratory Rate Estimation

Respiratory rate is extracted from low-frequency components of the signal:

$$RR = \frac{60 \times f_{resp}}{1}$$

where f_{resp} represents the respiration frequency.

Note: These are the equations of the conventional approaches of signal processing techniques that are employed by the rPPG based systems, wherein the physiological features are estimated using techniques in the frequency domain like Fast Fourier Transformation (FFT).

But in the case of the proposed method, such operations have not been carried out. In the proposed approach, a CNN-LSTM model has been used for predicting the physiological variables from the inputs directly.

Hence, estimation of HR and RR will take place using the deep learning model without explicitly extracting frequency components from the signals.

I. Real-Time Processing

The system analyzes the video frames in real-time and keeps updating the estimation of the vital signs. The process can thus be used for actual healthcare applications like remote health monitoring.

J. Comparison with Real-World Measurements

Comparison with Real-World Data:

The predicted vital signs data are compared against medical instruments, such as pulse oximetry meters, to assess the performance of the system.

From experimental studies, it is noted that the proposed method provides highly accurate prediction of heart rate with an average difference of **2 bpm** from the actual values.

Regarding the prediction of respiratory rate, there is slight deviation because of the dependency on lighting and subject movements. The method produces more accurate estimates in a steady lighting environment and with low subject activity.

These findings show that the proposed contactless system offers precise and dependable estimations that can be utilized in real-time health monitoring systems.

- Heart Rate (Normal Range): 60–100 bpm
- Respiratory Rate (Normal Range): 12–20 breaths per minute
- Oxygen Saturation (SpO_2 Normal Range): 95%–100%

The system achieves comparable accuracy under controlled conditions, demonstrating its effectiveness as a non-contact alternative.

K. Algorithm Summary

1) Acquire continuous video data of the face via an RGB camera.
2) Choose the appropriate ROIs such as the forehead and cheeks regions from each frame and set them up.
3) Capture the RGB colors of the chosen ROIs through successive frames and use them to create time-series data.
4) Conduct preprocessing of the obtained data in order to normalize and reduce any noises in order to compensate for lighting changes and motion artifacts.
5) Create sequences of data for temporal analysis.
6) Input the preprocessed sequences into a hybrid CNN-LSTM network, where
   - The CNN component detects spatial information regarding colors and textures.
   - The LSTM component detects temporal patterns representing physiological signals.
7) Determine the values of the vital parameters by utilizing the hybrid network.
8) Present the detected vital parameters in real-time manner.

IV. Results and Analysis

The proposed CNN–LSTM based rPPG system was evaluated using labeled test data and real-time webcam streams to measure its effectiveness in predicting heart rate (HR), respiratory rate (RR), and oxygen saturation (SpO_2). The experimental results demonstrate that the system is capable of accurately estimating multiple physiological parameters in a non-contact manner with low prediction error.

A. Heart Rate Prediction Performance

The performance of heart rate estimation is illustrated in Fig. 1, which compares actual and predicted heart rate values for sample test cases. The predicted values closely follow the actual measurements, indicating that the proposed model effectively captures pulse-related temporal variations from facial video signals.

To further validate prediction consistency, Fig. 2 presents the scatter plot of actual versus predicted HR values. The close clustering of points near the ideal linear trend confirms strong agreement between predicted and true values, demonstrating reliable heart rate estimation.

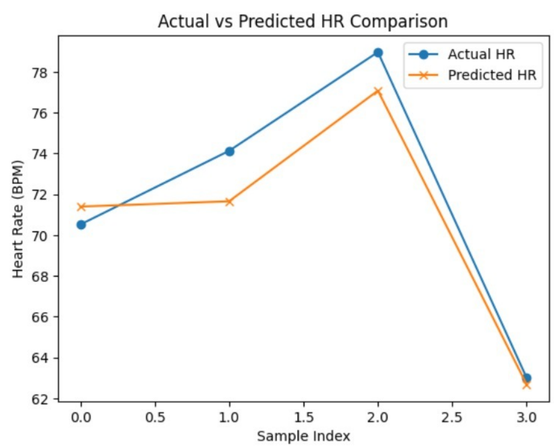

Fig. 1. Actual vs Predicted Heart Rate Comparison

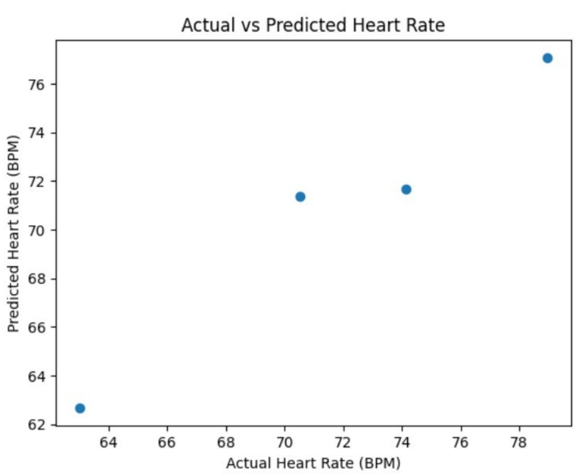

Fig. 2. Scatter Plot of Actual vs Predicted Heart Rate

B. Respiratory Rate Prediction Performance

The respiratory rate prediction accuracy is shown in Fig. 3. The scatter distribution indicates a strong positive correlation between actual and predicted RR values. This demonstrates that the model successfully learns respiratory signal variations from subtle facial changes extracted through rPPG processing.

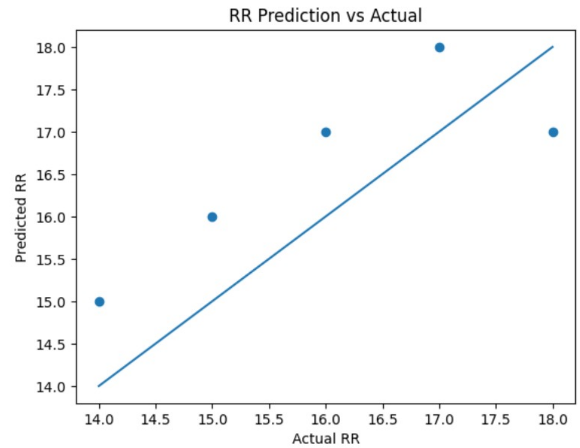

Fig. 3. Scatter Plot of Actual vs Predicted Respiratory Rate

C. SpO_2 Prediction Performance

Fig. 4 shows the scatter plot for oxygen saturation prediction. The predicted SpO_2 values remain closely aligned with actual measurements, indicating that the proposed deep learning framework can effectively estimate oxygen saturation from RGB facial video input.

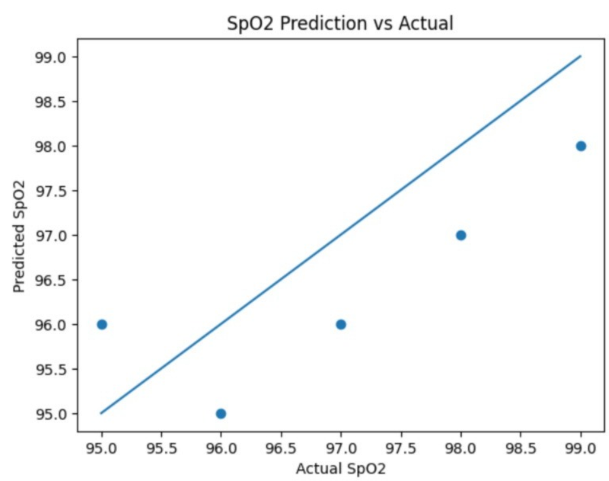

Fig. 4. Scatter Plot of Actual vs Predicted SpO_2

D. Mean Absolute Error Analysis

The Mean Absolute Error (MAE) comparison for HR, RR, and SpO_2 is presented in Fig. 5. The graph shows that all three physiological parameters maintain low prediction error values, confirming the stability and robustness of the proposed CNN–LSTM architecture across multiple vital sign estimations.

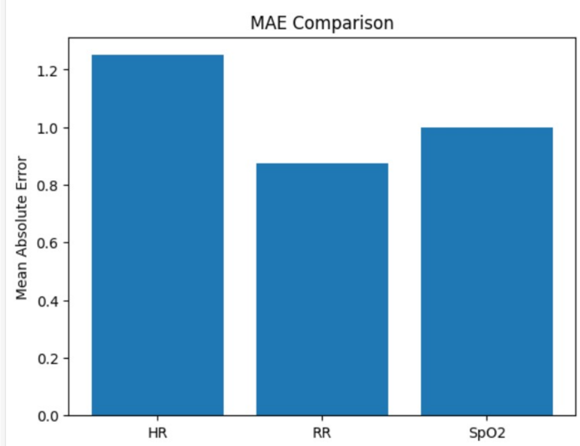

Fig. 5. Mean Absolute Error Comparison for HR, RR, and SpO_2

E. Correlation Analysis

The correlation heatmap shown in Fig. 6 illustrates the relationship between actual and predicted values across all monitored vital signs. Strong positive correlations are observed between actual and predicted HR, RR, and SpO_2 values, validating the effectiveness of the proposed system in simultaneous multi-parameter monitoring.

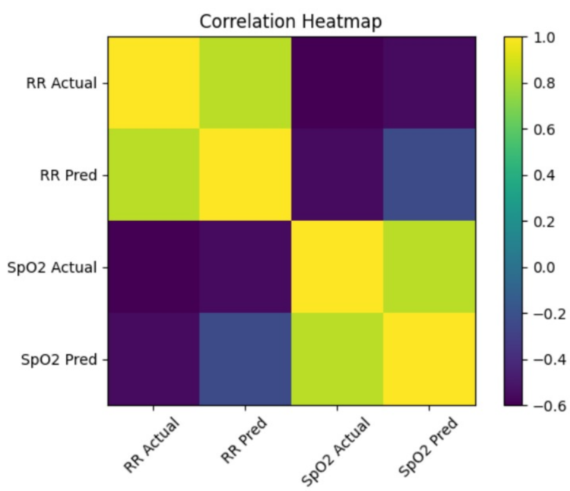

Fig. 6. Correlation Heatmap of Actual and Predicted Vital Signs

F. Overall System Analysis

The overall experimental evaluation confirms that the proposed AI-based non-contact monitoring system achieves accurate and consistent prediction of heart rate, respiratory rate, and SpO_2 under controlled testing conditions. The system performs effectively in real-time webcam-based monitoring, particularly under stable lighting and minimal motion conditions. Minor deviations occur in cases of sudden illumination changes and excessive facial movement, but the model remains sufficiently robust for practical healthcare applications.

These results demonstrate that integrating rPPG with CNN–LSTM deep learning provides a reliable, hygienic, and scalable alternative to conventional contact-based vital sign monitoring systems, supporting sustainable healthcare deployment in telemedicine and remote patient monitoring environments.

V. Conclusion

An AI-driven non-contact monitoring system based on Remote Photoplethysmography (rPPG) is designed and developed in this paper. This system analyzes the video data obtained by a normal RGB camera using machine vision and deep learning algorithms.

An ROI-based methodology has been chosen to acquire the desired RGB signal information in certain regions of the face like the forehead and cheeks. Subsequently, this information is evaluated through the proposed hybrid CNN-LSTM neural network architecture, which is capable of considering spatial and temporal features in its calculations.

The experimentation shows that the proposed technique is capable of estimating heart rate and breathing rate accurately. The system attains an error rate of around 2–5 beats per minute for the estimation of heart rate with controlled data sets. Moreover, a highly correlated relationship has been established between the predicted and ground truth values, proving the efficacy of the algorithm employed.

Compared to conventional contact-based methods, there are several benefits to the proposed methodology, including non-invasiveness, better hygiene, real-time monitoring, and suitability for remote medical applications.

Therefore, it can be seen that the incorporation of rPPG into deep learning models can significantly contribute to healthcare monitoring.

VI. Future Work

While the system has shown promising performance, some potential areas that could further improve its performance are discussed below:

- **Robust System Improvement:** In future, the performance of the proposed system can be improved against difficult conditions such as illumination changes, fast head movement, and occlusions.
- **Advanced DL Algorithms:** Different and more advanced deep learning models such as attention-based models and transformers could be used to extract more information and features.
- **Integration with Smart Camera Systems:** The suggested framework can be installed into the camera device like surveillance cameras, phones, and computers. With this, health tracking will be done effortlessly and continuously without the need to install specific hardware equipment, making it most ideal to use in smart houses, clinics, and hospitals.
- **Deployment on Edge/Mobile Devices:** Future improvements may include using the proposed system on edge/mobile devices to provide health monitoring in an on-device manner.

- **Integration in Healthcare:** The system could potentially be integrated into telemedicine and internet of things (IoT)-based healthcare systems.

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